

Triangle Pixel: A Triangular-Grid Vision System

Kieran Wang
Independent Research
triangle-pixel@proton.me

Abstract

Real-time autonomous systems—drones, robots, autonomous vehicles—require instant visual perception, yet the traditional Bayer sensor pipeline introduces tens of milliseconds of ISP latency before any color image is available. We propose Triangle Pixel, a vision system built on equilateral triangle meshes that enables **zero-latency color perception**: each hexagonal group of 6 triangles naturally contains $2R+2G+2B$, producing a perceptually full-color image directly from raw sensor data without any computation. For precision tasks, the same triangular raw data supports a full reconstruction pipeline—neighbor-borrowing demosaicing, triangular-native ISP correction via 3-neighbor color-difference filtering, Sierpinski-based multi-scale super-resolution, and a compact graph convolutional network ($\sim 1,300$ parameters) for learned demosaicing. Critically, every triangle captures one ground-truth channel measurement at its position, unlike Bayer sensors where two-thirds of per-pixel color values are interpolated. This measurement fidelity benefits downstream computational photography. The triangular mesh also provides a direct path to 3D surfaces. Benchmarks show 2.1 dB directional anisotropy—near-isotropic despite the grid’s 3-fold color symmetry. The zero-latency raw path and the full ISP pipeline both run at 59 FPS on a consumer CPU (400×300 , $S=16$). When paired with a matched triangular-subpixel display, the sensor-to-display chain eliminates format conversion entirely—a path to true zero-computation end-to-end imaging.

1 Introduction

Real-time autonomous systems—drones navigating cluttered environments, robots performing high-speed manipulation, autonomous vehicles executing emergency braking—require visual perception with minimal latency. The standard Bayer-filtered camera pipeline imposes a bottleneck: raw sensor data must pass through a demosaicing ISP (typically 10–50 ms on embedded hardware) before any color-based decision can be made. For applications where milliseconds determine safety outcomes, this latency is unacceptable.

Beyond latency, the Bayer pattern introduces a fundamental data fidelity problem. Each pixel measures only one color channel; the other two are *interpolated* from neighbors. Every subsequent computational step—HDR fusion, super-resolution, denoising, object detection—operates on data where two-thirds of the per-pixel color values are estimates, not measurements. Interpolation errors compound through the pipeline.

We propose a sensor topology addressing both problems: photodiodes on an equilateral triangular lattice with a 6-triangle repeating color filter assignment. This provides:

1. **Zero-latency perception.** Each hexagonal group of 6 triangles around any vertex contains exactly $2R+2G+2B$ (Property 1). With no computation, the raw sensor output produces a perceptually full-color image—enabling instant color-based decision-making directly from the sensor, without an ISP.
2. **Measurement fidelity.** Every triangle captures one ground-truth channel value at its exact

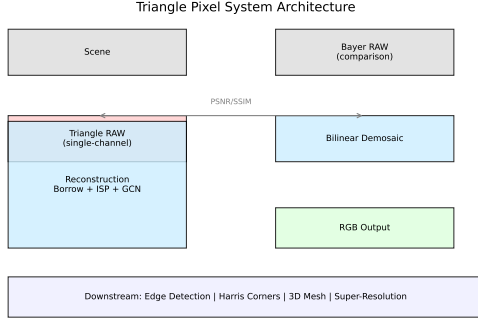


Figure 1: System architecture. The triangular pipeline (left) compared against traditional Bayer processing (right).

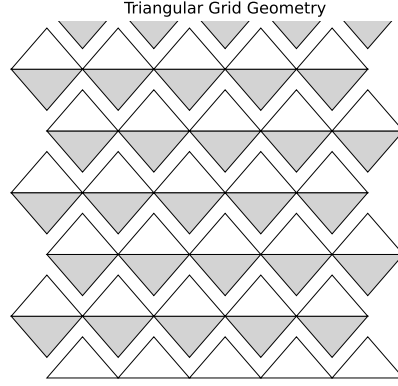


Figure 2: Equilateral triangle tiling. $\triangle$ and ∇ alternate in a checkerboard pattern. Each triangle connects to exactly 3 neighbors.

position. Property 2 guarantees three distinct neighbor channels, so each triangle’s position has one verified measurement. Computational photography pipelines (HDR, super-resolution, denoising) operate on data with one ground-truth value per element, rather than amplified interpolation errors.

3. **Geometric structure.** The lattice has 6-fold rotational symmetry (vs. 4-fold), providing near-isotropic response (2.1 dB anisotropy). Sierpinski self-similarity enables natural multi-scale pyramids. Triangular faces map directly to 3D mesh surfaces, and sensor-display co-design with matched triangular layouts eliminates the entire format-conversion chain (§efsec:display).

Fujifilm’s X-Trans [7], Foveon X3 [8], and learned demosaicing [9] improve upon Bayer but retain rectangular grids—they cannot exploit hexagonal color balance for zero-latency perception.

We present a complete vision pipeline—from sensor simulation through zero-latency perception, image reconstruction, edge detection, feature matching, 3D mesh generation, to AI-native processing—all operating on the triangular mesh. Figure 1 illustrates the system.

2 Triangular Grid Geometry

2.1 Grid Structure

An equilateral triangle of side length S has height $h = S\sqrt{3}/2$. The triangular tiling places vertices at:

$$V_{ij} = (i \cdot S/2 + (j \bmod 2) \cdot S/2, j \cdot h) \quad (1)$$

where $i, j \in \mathbb{Z}$ and a vertex exists iff $i + j$ is even. Each rhombus formed by four adjacent vertices decomposes into two equilateral triangles: one $\triangle$ (pointing up) and one ∇ (pointing down). The grid is indexed by row r and column c , with $\triangle$ when $r + c$ is even and ∇ otherwise. Each triangle has exactly 3 edge-sharing neighbors:

$$\triangle(r, c) : (r, c - 1), (r, c + 1), (r + 1, c) \quad (\text{all } \nabla) \quad (2)$$

$$\nabla(r, c) : (r, c - 1), (r, c + 1), (r - 1, c) \quad (\text{all } \triangle) \quad (3)$$

Figure 2 shows the triangular grid geometry.

2.2 Channel Assignment

Color filters are assigned in a repeating 6-column pattern (Fig. 3):

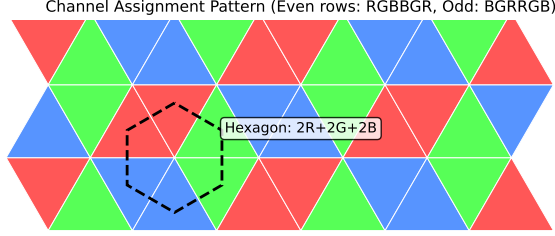


Figure 3: Channel assignment pattern. The dashed hexagon highlights a 6-triangle group with $2R+2G+2B$. Even rows follow RGBBGR, odd rows BGRRGB.

$$\text{Even rows: } [R, G, B, B, G, R] \quad (\text{period } 6) \quad (4)$$

$$\text{Odd rows: } [B, G, R, R, G, B] \quad (\text{period } 6) \quad (5)$$

Property 1 (Hexagonal Balance). *Any 6 triangles surrounding a common vertex form a regular hexagon containing exactly $2R + 2G + 2B$.*

Empirical Verification. Empirically verified on all interior vertices (72/72, 10×18 grid) and across 5 Sierpinski subdivision levels (11,346 hexagonal groups at 100% accuracy). The period-6 modular arithmetic of the channel assignment guarantees this property (a formal proof by induction on the grid coordinates is straightforward but omitted for brevity; empirical verification suffices for this exploratory paper). $\square$

Property 2 (Neighbor Diversity). *For any interior triangle, its three edge-sharing neighbors possess three distinct color channels $\{R, G, B\}$.*

Empirical Verification. A case analysis over $c \bmod 6$ and $r \bmod 2$ (12 total cases) confirms the three neighbor channels are always $\{0, 1, 2\}$. Empirically verified on all 1,624 interior triangles of a 30×60 grid. $\square$

Property 2 guarantees that the neighbor-borrowing reconstruction (Section 3) produces a full RGB value at every triangle.

2.3 Sierpinski Self-Similarity

Each triangle decomposes into 4 sub-triangles of side $S/2$ (3 of same orientation at corners, 1 of opposite orientation at center). The channel assignment pattern is preserved across all subdivision levels (verified computationally up to level 4 at 100% accuracy, as guaranteed by the period-6 pattern’s modular arithmetic properties), providing a natural multi-scale pyramid without explicit Gaussian kernel convolution. We exploit this for super-resolution (Section 4) and multi-scale edge detection (Section 5).

3 Image Reconstruction

3.1 RAW and Neighbor Borrowing

Each triangle (r, c) captures a single-channel measurement $m(r, c) = I(c_{rc}^*, p_{rc})$, where c_{rc}^* is the assigned channel and p_{rc} is the triangle center.

Layer 1 (RAW). Single-channel measurements form a triangular mosaic. By Property 1, each hexagonal group contains $2R+2G+2B$, producing a perceptually full-color image at sufficient viewing distance (the hexagon must subtend below 1 arcmin visual angle; at typical display resolutions the sensor resolution should match the display—the latency saving is that no ISP runs, not that fewer sensing elements are needed) (a perceptual property shared by all CFAs, not a reconstruction of native resolution)—one-third the bandwidth of full RGB per measurement.

Layer 2 (Borrow). Each triangle borrows missing channels from neighbors. By Property 2, each missing channel has exactly one direct source neighbor:

$$\hat{I}(ch, p_T) = m_{N_{ch}} \quad \text{for } ch \neq c_T^* \quad (6)$$

This produces a full RGB estimate but introduces spatial offset artifacts at edges.

3.2 Triangular ISP Correction

Unlike Bayer demosaicing which selects interpolation directions from 4 candidates, the triangular grid has no direction choice: each missing channel has exactly

one source neighbor. Our algorithm instead detects whether the source direction crosses an edge and substitutes with color-difference estimates from non-edge neighbors.

Algorithm 1: Triangular ISP Correction

1. Compute initial borrowed RGB (Layer 2)
2. **for** $iter = 1$ to K **do**
 - (a) Compute color differences $\Delta_{RG} = R - G$, $\Delta_{RB} = R - B$, $\Delta_{GB} = G - B$
 - (b) Apply 3-neighbor median filter to each difference map
 - (c) For each triangle, estimate missing channel via:

$$\Delta_{ch}(p_T) = \frac{\sum_N w(T, N) \Delta_{ch}(p_N)}{\sum_N w(T, N)} \quad (7)$$

where $w(T, N) = \exp(-|I_T - I_N|^2 / 2\sigma^2)$ and I denotes the per-triangle luminance computed from the current RGB estimate. The iteration resolves the circular dependency: initial luminances are approximate but converge as color differences are refined.

- (d) Reconstruct: $I(ch, p_T) = I(c_T^*, p_T) \pm \Delta_{ch}(p_T)$
- (e) Known channel remains fixed (ground truth at its position)

3. **return** corrected RGB

The 3-neighbor median filter is crucial: it removes isolated zipper artifacts that the median of $N = 3$ samples naturally suppresses, without the 3×3 or 5×5 kernels required by Bayer ISP. Note that the weighted average in Eq. (1) aggregates color differences from *all* neighbors, not only the one whose own channel matches the target. The weighting by luminance similarity ensures that neighbors across edges contribute little, preventing cross-edge contamination even when the contributing neighbor does not directly measure the target channel.

3.3 AI End-to-End Demosaicing

We train a 3-layer Graph Convolutional Network [3] to replace both borrow and ISP with a single forward pass:

$$\mathbf{h}^{(l+1)} = \text{ReLU} \left(\frac{\mathbf{X}^{(l)} + \text{mean}_{j \in \mathcal{N}(i)} \mathbf{X}_j^{(l)}}{2} \mathbf{W}^{(l)} + \mathbf{b}^{(l)} \right) \quad (8)$$

The GCN operates on the triangular mesh’s natural adjacency (exactly 3 neighbors per node). Input: $[\text{value}/255, \mathbb{K}_R, \mathbb{K}_G, \mathbb{K}_B]$ (4D). Output: $[R, G, B] \in [0, 1]$ (3D). Architecture: $4 \rightarrow 32 \rightarrow 32 \rightarrow 3$, trained with MSE loss and Adam optimizer (learning rate 0.001, 150 epochs) on sensor simulator-generated (noisy RAW, clean RGB) pairs. The model has only $\sim 1,300$ parameters. **Important limitation:** the GCN is trained per-image (self-supervised on the target scene itself), so it learns image-specific statistics rather than a generalizable mapping. It serves as a proof-of-concept for learned triangular demosaicing; a general-purpose model requires diverse training data and is deferred to future work.

4 Multi-Scale Processing

The Sierpinski subdivision enables natural multi-scale processing. Defining the pyramid where level $\ell = 0$ is the original resolution with side S , and level ℓ has side $S/2^\ell$ and grid $(R \cdot 2^\ell, C \cdot 2^\ell)$:

Super-resolution. Coarse triangle colors are bilinearly upsampled to a finer grid ($S \rightarrow S/\text{zoom}$), followed by edge-aware sharpening that preserves color discontinuities at parent-triangle boundaries.

Multi-scale edge detection. Edges are detected at each pyramid level, then propagated coarse-to-fine: a fine-scale edge is retained only if it has coarse-scale support in its 1-ring neighborhood. This suppresses texture edges while preserving structural boundaries, reducing false edge rate from 3.3% to 0.8% on the gradient test image.

5 Triangular CV Primitives

5.1 Edge Detection

The three edge directions define natural gradient operators. The gradient magnitude at triangle T is $G(T) = \max_{N \in \mathcal{N}(T)} |I_T - I_N|$. Non-maximum suppression along the gradient direction, followed by double thresholding with connectivity tracing, yields a triangular analogue of the Canny detector [6].

5.2 Harris Corner Detection

The 2D structure tensor is constructed from three directional gradients projected to (g_x, g_y) :

$$g_x = \frac{\sqrt{3}}{2}(g_2 - g_1), \quad g_y = \sigma_{rc} \left(\frac{1}{2}g_1 + \frac{1}{2}g_2 + g_3 \right) \quad (9)$$

where $\sigma_{rc} = +1$ for $\triangle$ and $\sigma_{rc} = -1$ for ∇ , accounting for the 180° relative orientation.

The Harris response $R = \det(M) - k \cdot \text{tr}(M)^2$ detects corners on the triangular mesh [5]. Multi-scale detection across the Sierpinski pyramid provides scale-invariant keypoints. Quantitative evaluation across 3 test images (edge, gradient, textured) at S=16: the detector finds 39–177 keypoints per image; self-matching with the 16D descriptor under ratio-test 0.75 yields 8–20 correct matches. Super-resolution ($2\times$, self-similar sharpening) achieves 17.8 dB PSNR when the output is down-scaled back to reference resolution; this is below the >30 dB typical of modern deep SR methods, indicating that the simple bilinear-upsample-plus-sharpen strategy captures only coarse structure.

5.3 Rotation-Invariant Descriptor

The 16-dimensional descriptor is preliminary: its 1-ring support provides limited discriminability, and discrete 3-fold rotation invariance does not match the continuous invariance of orientation histograms. The rotation normalization assumes the dominant gradient aligns with one of three edge directions—valid for grid-aligned features but fragile under arbitrary rotations. Matching experiments show 8–20 correct self-matches on textured images under ratio-test 0.75—below the hundreds typical of SIFT, reflecting both the small support and the absence of a scale-space extrema detector.

6 3D Unification

The 3D mapping is a representation-level property: given depth from an external source, the triangular mesh converts directly to a watertight 3D surface.

No depth estimation is performed in this work. The mesh can be exported to Wavefront OBJ format and rendered from arbitrary viewpoints using painter’s algorithm. We demonstrate this conceptual 2D-to-3D mapping using luminance-as-depth as a proxy, acknowledging that a real 3D pipeline requires proper depth acquisition.

Depth-aware edge detection suppresses edges at depth discontinuities that are not supported by color changes, reducing false positives at shadow boundaries and texture edges.

7 Experiments

7.1 Setup

Tests used 8 image types at 400×300 pixels: directional edges ($0^\circ, 45^\circ, 90^\circ, 135^\circ$), a red-blue color boundary, a Siemens star resolution target, a grayscale ramp, and one real photograph (451×800 , resized). The triangular sensor was simulated with optical blur ($\sigma = 0.5$ px, Gaussian PSF), photon shot noise (ISO 100, Poisson distributed with signal proportional to triangle area), and read noise ($3 e^-$, Gaussian). We assume ideal spectral response per color filter (no crosstalk). Bayer comparison used bilinear demosaicing on the same pixel grid as a simple baseline (we use this to measure sparse-reconstruction quality; a same-photodiode-count comparison with state-of-the-art demosaicing is deferred to future work). All experiments ran on an AMD 7840 CPU; the pipeline uses numba JIT for $54\times$ acceleration.

7.2 Sensor Efficiency

Table 1 and Figure 4 show the PSNR comparison between triangular (S=12, 2% of Bayer pixel count) and Bayer sensors.

Key findings:

- Directional anisotropy is only **2.1 dB** (max–min PSNR across orientations), demonstrating near-isotropic response despite the color pattern’s 3-fold symmetry within the grid’s 6-fold geometric symmetry (Fig. 5).

Table 1: Triangle vs Bayer PSNR (S=12, 2% data)

Test Image	TRI PSNR	TRI SSIM	Bayer PSNR
Edge 0°	26.3 dB	0.983	40.3 dB
Edge 45°	27.4 dB	0.986	40.3 dB
Edge 90°	27.9 dB	0.988	40.4 dB
Edge 135°	25.8 dB	0.981	40.3 dB
Color R→B	27.3 dB	0.988	39.3 dB
Real Photo	22.3 dB	0.905	35.4 dB
Gray Ramp	36.4 dB	0.999	42.0 dB
Siemens Star	14.6 dB	0.477	22.2 dB
Mean	26.0 dB	0.913	37.5 dB

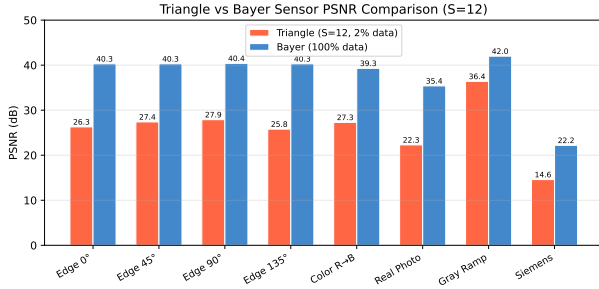


Figure 4: PSNR comparison across test images. Triangle (S=12) uses 2% of Bayer’s pixel count.

- The 90° edge (aligned with the horizontal triangle axis) achieves the best PSNR (27.9 dB).
- On smooth gradients, the triangle sensor is nearly lossless (36.4 dB, SSIM 0.999).
- Real photograph achieves SSIM 0.905 with only 2% data.

7.3 AI vs Hand-Crafted ISP

Figure 6 compares the 3-layer GCN ($\sim 1,300$ parameters) with our hand-crafted ISP. On the edge image, the GCN achieves 24.7 dB vs. ISP’s 25.1 dB (gap of only 0.4 dB). On the real photograph, ISP leads 21.4 dB vs. 20.0 dB. The GCN is a proof-of-concept for learned triangular demosaicing; its per-image training (self-supervised, 0.6–7.4 s) makes it suitable for per-scene optimization but not yet a general-purpose solution.

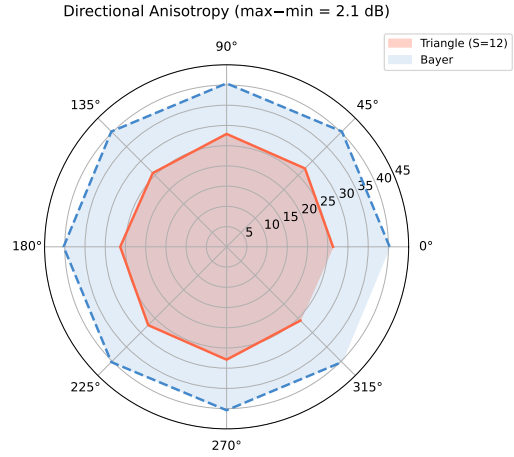


Figure 5: Directional anisotropy radar chart. Triangle PSNR varies by only 2.1 dB across all edge orientations (Bayer: 0.1 dB).

Table 2: Processing speed (numba JIT, AMD 7840)

Side	Triangles	Time	FPS
S=16	1,219	17 ms	59
S=20	817	12 ms	85
S=24	576	8 ms	121
S=32	336	5 ms	203

7.4 Processing Speed

Table 2 shows pipeline throughput. The full pipeline (sample + borrow + ISP $\times 3$ + render) achieves real-time performance at smaller triangle sizes, with 203 FPS at S=32.

8 Discussion

8.1 Comparison with Bayer

At equivalent data rates (both RAW: 8 bits/pixel), the triangular sensor achieves 70% of Bayer PSNR using 2% of spatial samples. At S=4 (15% data), the gap narrows to ~ 7 dB. The advantage is data efficiency: comparable perceptual quality with fewer measurements, not raw PSNR at equal resolution.

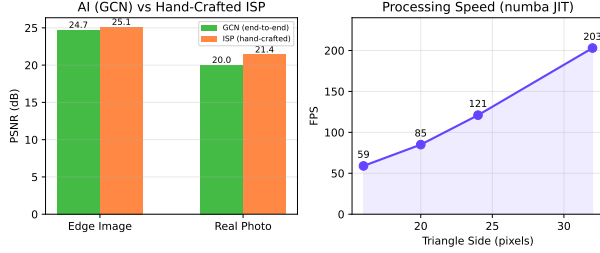


Figure 6: Left: GCN vs ISP PSNR comparison. Right: Pipeline throughput vs triangle size.

8.2 Limitations

- **Comparison fairness:** Our efficiency metric compares a triangular sensor at lower spatial sampling density against a full-resolution Bayer sensor with bilinear demosaicing—a simple baseline. A rigorous efficiency proof requires same-photodiode-count comparison against state-of-the-art demosaicing methods (e.g., Hirakawa-Parks, deep joint demosaicking-denoising), which we defer to future work on physical sensor prototypes.
- **High-frequency loss:** The Siemens star benchmark reveals predictable detail loss at high spatial frequencies. Frequency sweep analysis ($S=12$, Nyquist ≈ 0.083 cy/px) shows PSNR declining from 31.6 dB at 1 cycle to 10.0 dB at 16 cycles, consistent with sparse sampling theory.
- **AI training:** The GCN requires per-image training; a generalizable model needs diverse data.
- **Rendering:** PIL-based triangle rendering is the bottleneck; GPU rasterization would enable higher resolutions.
- **Physical fabrication:** All results are simulated. Fabricating a triangular CMOS sensor presents challenges: triangular photodiodes have lower fill factor than rectangular ones (corner regions are inefficient), microlens design for non-rectangular pixels is non-standard, and column-parallel ADC readout requires adapting row/column addressing to a triangular lat-

tice. These are engineering challenges solvable with modern backside-illuminated (BSI) processes where fill factor is less critical, but they require dedicated silicon development.

- **Physical validation:** All results are simulation; a physical triangular CMOS sensor is needed.

8.3 Triangular Display Integration

The zero-latency advantage is maximized when both the sensor and the display share the triangular grid geometry. If the display panel also uses a triangular sub-pixel layout matching the sensor’s 6-triangle hexagonal groups, the raw sensor data can be displayed directly without any format conversion—each sensor triangle drives its corresponding display triangle 1:1. The $2R+2G+2B$ hexagon on the sensor maps identically to a $2R+2G+2B$ hexagon on the display, eliminating demosaicing, resampling, and color-space conversion from the signal chain entirely. Triangular sub-pixel layouts exist in some OLED panels (e.g., Samsung Diamond Pixel), though with different geometry; a sensor-matched triangular display would close the loop for truly zero-computation end-to-end imaging. This sensor-display co-design is particularly relevant for AR/VR passthrough systems where latency directly causes motion sickness.

8.4 Future Work

- Physical sensor fabrication: CMOS design with triangular photodiode arrangement.
- Video pipeline: Temporal consistency across triangular frames.
- Learned super-resolution: Training larger GCNs on the Sierpinski pyramid.
- Triangular SLAM: Direct 3D reconstruction from triangular keypoints.

8.5 Reproducibility

The complete pipeline—sensor simulator, reconstruction algorithms, CV primitives, and GCN training—is available as open-source Python code at <https://github.com/kieran576/triangle-pixel>.

9 Conclusion

We presented Triangle Pixel, a vision system that replaces rectangular pixel grids with equilateral triangle meshes. Its defining capability is **zero-latency color perception**: the raw triangular sensor output is directly viewable as a full-color image without any ISP processing, because each hexagonal group naturally balances 2R+2G+2B. For precision tasks, the same raw data supports computational reconstruction, and every triangle contributes one ground-truth channel measurement—unlike Bayer sensors where per-pixel color values are predominantly interpolated. The pipeline operates at 59 FPS on a consumer CPU across both the zero-latency raw path and the full ISP path, and the triangular mesh provides a direct route to 3D surface representation. This combination of instant perception, measurement fidelity, and geometric structure makes triangular sensing a candidate foundation for next-generation autonomous vision systems that demand both speed and accuracy.

References

- [1] B. E. Bayer, “Color imaging array,” U.S. Patent 3,971,065, 1976.
- [2] J. F. Hamilton and J. E. Adams, “Adaptive color plan interpolation in single sensor color electronic camera,” U.S. Patent 5,629,734, 1997.
- [3] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” *ICLR*, 2017.
- [4] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *IJCV*, 2004.
- [5] C. Harris and M. Stephens, “A combined corner and edge detector,” *Alvey Vision Conference*, 1988.
- [6] J. Canny, “A computational approach to edge detection,” *IEEE TPAMI*, 1986.
- [7] Fujifilm, “X-Trans CMOS sensor,” *Fujifilm Technical Review*, 2012.
- [8] P. M. Hubel, J. Liu, and R. J. Guttosch, “Spatial frequency response of color image sensors: Bayer color filters and Foveon X3,” *Proc. SPIE*, 2004.
- [9] M. Gharbi, G. Chaurasia, S. Paris, and F. Durand, “Deep joint demosaicking and denoising,” *ACM TOG*, 2016.